Solution:

Moment of inertia of a solid cylinder about its axis $I_s = \frac{1}{2}MR^2$

Moment of inertia of a hollow cylinder about its axis $I_h = MR^2$

$$I_s = \frac{1}{2}I_h \text{ or } I_h = 2I_s$$

$$torque\ \tau = I\alpha$$

$$\alpha = \frac{\tau}{I}$$

$$\alpha_s = \frac{\tau}{I_s} \text{ and } \alpha_h = \frac{\tau}{I_h}$$

$$\alpha_s I_s = \alpha_h I_h \Rightarrow \alpha_s = \alpha_h \frac{I_h}{I_s}$$

Since, $I_h > I_s \Rightarrow \frac{I_h}{I_s} > 1$

$$\therefore \alpha_s > \alpha_h$$

For the same torque, a solid cylinder gets more acceleration than a hollow cylinder.

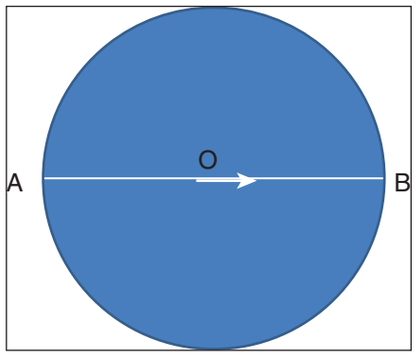
Note: The above two cylinders must be made up of materials of different density. (Say why?)

10. **A thin horizontal circular disc is rotating about a vertical axis passing through its centre. An insect goes from A to point B along its diameter as shown in Figure. Discuss how the angular speed of the circular disc changes?**

Solution:

As the disc is freely rotating, with the insect on it, the angular momentum of the system is conserved.

$$L = I\omega = \text{constant}$$



When the insect moves towards the centre (from A to O), the moment of inertia (I) decreases. Thus, the angular velocity (ω) increases. When it moves away from centre (from O to B), the moment of inertia (I) increases. Thus, the angular velocity (ω) decreases.

11. (i) What is the shape of the graph between $\sqrt{E_{kr}}$ and L? (E_{kr} is the rotational kinetic energy and L is angular momentum)

(ii) What information can you get from the slope of the graph?

(iii) You are given the graph of $\sqrt{E_{kr}}$ and L for two bodies A and B. Which one has more moment of inertia?

Solution:

i) We know that, Rotational kinetic Energy

$$E_{kr} = \frac{1}{2}I\omega^2$$

$$= \frac{1}{2}I\omega \times \omega = \frac{1}{2}.L.\omega = \frac{1}{2}\frac{L^2}{I} \qquad \because L = I\omega,\ \omega = L/I$$

$$E_{kr} = \frac{L^2}{2I}$$

$$L^2 = 2IE_{kr}$$

$$L = \sqrt{2IE_{kr}} = \sqrt{2I}.\sqrt{E_{kr}}$$

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